

ICSE CLASS IX EXAMINATION

MATHEMATICS

Model Question Paper

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

- Answers to this Paper must be written on the answer sheet provided separately.
- Section A (40 Marks):** Compulsory. Attempt all questions.
- Section B (40 Marks):** Attempt any **four** questions out of six.
- All working, including rough work, must be clearly shown alongside the answers.

SECTION A (40 MARKS) — COMPULSORY

Question 1

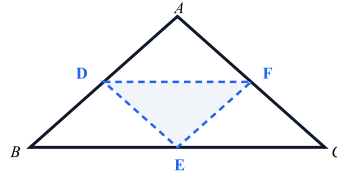
[10 × 1 = 10]

Choose the correct option for each of the following questions:

- (i) The simplest rationalising factor of $\frac{1}{\sqrt{7} - \sqrt{3}}$ is:
- (a) $\sqrt{7} - \sqrt{3}$ (b) $\sqrt{7} + \sqrt{3}$
(c) $7 + 3$ (d) $\sqrt{21}$
- (ii) If $x + \frac{1}{x} = 5$, then the value of $x^2 + \frac{1}{x^2}$ is:
- (a) 23 (b) 25
(c) 27 (d) 21
- (iii) The factors of $x^2 - 11x + 28$ are:
- (a) $(x - 7)(x - 4)$ (b) $(x + 7)(x - 4)$
(c) $(x - 7)(x + 4)$ (d) $(x + 7)(x + 4)$
- (iv) The linear equations $3x + 2y = 12$ and $x - y = 1$ intersect at the point:
- (a) (2, 3) (b) (3, 2)
(c) (1, 4) (d) (4, 1)
- (v) In a right-angled triangle ABC right-angled at B , if $AB = 9\text{ cm}$ and $BC = 12\text{ cm}$, then length of AC is:
- (a) 13 cm (b) 14 cm
(c) 15 cm (d) 21 cm
- (vi) In $\triangle ABC$, D and E are midpoints of AB and AC respectively. If $BC = 14\text{ cm}$, then length of DE is:
- (a) 28 cm (b) 7 cm
(c) 14 cm (d) 3.5 cm
- (vii) Which of the following cannot be the length of the third side of a triangle whose two sides are 6 cm and 8 cm ?
- (a) 3 cm (b) 11 cm
(c) 13 cm (d) 15 cm
- (viii) The expansion of $(2a - 3b)^3$ contains the coefficient of a^2b equal to:
- (a) -36 (b) +36
(c) -54 (d) +54
- (ix) If $\sqrt{2} = 1.414$, the value of $\frac{1}{\sqrt{2}}$ correct to two decimal places is:
- (a) 0.71 (b) 1.41
(c) 0.50 (d) 0.35
- (x) In $\triangle PQR$, if $\angle P = 50^\circ$ and $\angle Q = 70^\circ$, the longest side of the triangle is:
- (a) PQ (b) QR
(c) PR (d) Cannot be determined

Question 3

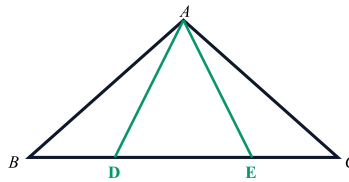
- (a) If $x - \frac{1}{x} = 4$, find the values of: (i) $x^2 + \frac{1}{x^2}$ (ii) $x^4 + \frac{1}{x^4}$ [4]
- (b) In $\triangle ABC$ shown below, D , E , and F are midpoints of AB , BC , and CA respectively. If perimeter of $\triangle ABC = 32 \text{ cm}$, calculate the perimeter of $\triangle DEF$. [3]



- (c) A ladder 13 m long reaches a window of a building 12 m above the ground. Calculate the distance of the foot of the ladder from the base of the wall. [3]

Question 4

- (a) Factorise using middle term splitting/grouping: $2x^3 - 3x^2 - 17x + 30$. [4]
- (b) In the given figure, $\triangle ABC$ is isosceles with $AB = AC$. Points D and E are on BC such that $BD = CE$. Prove that $AD = AE$. [3]

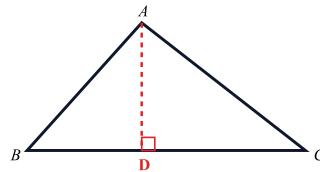


- (c) Solve for x and y : $\frac{(x+1)}{2} + \frac{(y-1)}{3} = 8$ and $\frac{(x-1)}{3} + \frac{(y+1)}{2} = 9$. [3]

SECTION B (40 MARKS) — ANSWER ANY FOUR QUESTIONS

Question 5

- (a) Find rational constants a and b if: $\frac{(\sqrt{5} + \sqrt{3})}{(\sqrt{5} - \sqrt{3})} = a + b\sqrt{15}$ [4]
- (b) Expand completely: $(2x - \frac{1}{x} + 3)^2$. [3]
- (c) In $\triangle ABC$ below, $AD \perp BC$. Using Pythagoras Theorem in $\triangle ABD$ and $\triangle ACD$, prove that $AB^2 - AC^2 = BD^2 - CD^2$. [3]

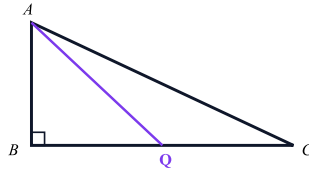


Question 6

- (a) Solve the pair of simultaneous equations algebraically and check consistency: $x + y = 6$ and $2x - y = 3$. [4]
- (b) Factorise: $x^2 + \frac{1}{x^2} - 2 - 3x + \frac{3}{x}$. [3]
- (c) In quadrilateral $ABCD$, E , F , G , and H are midpoints of sides AB , BC , CD , and DA respectively. Using the Midpoint Theorem, prove that quadrilateral $EFGH$ is a parallelogram. [3]

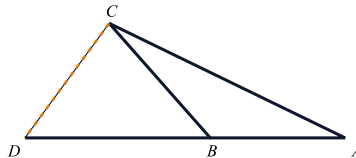
Question 7

- (a) Sum of two numbers is 45 and their difference is 9. Find the numbers using simultaneous linear equations. [4]
- (b) Without actual direct expansion, evaluate using algebraic identities: $(28)^3 + (-15)^3 + (-13)^3$. [3]
- (c) In $\triangle ABC$ right-angled at B , P and Q are the midpoints of sides AB and BC respectively. Apply Pythagoras Theorem to calculate and prove: $4AQ^2 = 4AB^2 + BC^2$. [3]



Question 8

- (a) Rationalise the denominator and simplify: $\frac{1}{1 + \sqrt{2} - \sqrt{3}}$. [4]
- (b) In $\triangle ABC$, side AB is produced to D such that $BD = BC$. If $\angle A = 60^\circ$ and $\angle B = 80^\circ$, calculate $\angle ACD$ and state which side of $\triangle ADC$ is the longest. [3]



- (c) Factorise: $(x^2 - 3x)^2 - 8(x^2 - 3x) - 20$. [3]

Question 9

- (a) Person A and Person B together have ₹1200. If A gives ₹100 to B, then B will have twice as much money as left with A. [4]
How much money did each have originally?
- (b) In $\triangle ABC$, D and E are midpoints of AB and AC . If $DE = (2x + 1)$ cm and $BC = (5x - 4)$ cm, calculate the value of x and [3]
length of BC using Midpoint Theorem properties.
- (c) If $x^2 + \frac{1}{x^2} = 14$, find the value of $x^3 + \frac{1}{x^3}$ (take positive value of $x + \frac{1}{x}$). [3]

Question 10

- (a) In a right-angled triangle ABC , $\angle B = 90^\circ$. If $AB = x$ cm, $BC = (x + 7)$ cm, and hypotenuse $AC = (x + 8)$ cm, calculate [4]
the value of x and find the perimeter of $\triangle ABC$.
- (b) Simplify the expression using expansion identities: $\frac{[(a + b)^3 - (a - b)^3]}{2b}$. [3]
- (c) If $x = \frac{(\sqrt{3} + 1)}{(\sqrt{3} - 1)}$ and $y = \frac{(\sqrt{3} - 1)}{(\sqrt{3} + 1)}$, find the value of $x^2 + y^2$. [3]